

BST Physics Curriculum Vol 14 Chapter 6 — Bell Sub-Tsirelson Information Bound v0.4 (refilled per Cal #104)

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Vol 14 Chapter 6 — Bell Sub-Tsirelson Information Bound

Chapter motivation

Bell's inequality (Bell 1964): for any local-hidden-variable theory, the CHSH (Clauser-Horne-Shimony-Holt) quantity $|S| \leq 2$. Standard QM violates this: $|S|_{\text{QM}} \leq 2\sqrt{2}$ (Tsirelson 1980 bound; $S^2 \leq 8$). Aspect 1982 + Hensen 2015 + ongoing precision experiments confirm QM violation; Bell inequality + Tsirelson bound foundational results in quantum foundations.

BST predicts sub-Tsirelson deviation $1/2^{\wedge}N_c = 1/8 = 0.125$: T2399 substrate Bell-CHSH gives $\text{Tr}(B^2) = 126/16 = 7.875$ (Calibration #17 Wednesday trace-level integrated-capacity framing); $\text{Tsirelson}^2 - \text{Tr}(B^2) = 8 - 126/16 = 1/8 = 1/2^{\wedge}N_c$ (substrate signature at depth $N_c = 3$ below Tsirelson). T2469 SCMP framework (Friday Casey-named #8 STANDING): substrate-mediated observer correlations exhibit this $1/8$ deviation as Layer 1 operational claim.

Information-theoretic interpretation: Bell-CHSH bounds quantify information-extractable from quantum correlations; substrate sub-Tsirelson $1/8$ IS substrate-mediation signature distinguishing BST from standard QM channel capacity (Vol 14 Ch 3).

Section 6.0b — Reader-grade 3-level pedagogy

Level 1: Bell-CHSH inequality $|S| \leq 2$ (local-hidden-variable), $|S|_{\text{QM}} \leq 2\sqrt{2}$ (Tsirelson 1980 QM bound); BST T2399 + Cal #17: $\text{Tr}(B^2) = 126/16$ EXACT trace-level; sub-Tsirelson $1/2^{\wedge}N_c = 1/8 = 0.125$ substrate signature; SP-30 Bell experiment design \$300-500K operational falsifier; information-theoretic interpretation: substrate-mediation distinguishes BST channel capacity.

Level 2 (graduate-physicist/information-theorist): Bell's inequality (John Bell 1964 *On the Einstein-Podolsky-Rosen Paradox*, Physics 1): for any local-hidden-variable (LHV) theory, the CHSH (Clauser-Horne-Shimony-Holt 1969 *Proposed Experiment to Test Local Hidden-Variable Theories*, PRL 23) quantity $|S| \leq 2$ where $S = \langle A_1 B_1 \rangle + \langle A_1 B_2 \rangle + \langle A_2 B_1 \rangle - \langle A_2 B_2 \rangle$ involves four 2-outcome measurements (A_i Alice's, B_j Bob's). Standard QM violates Bell: maximum $|S|_{\text{QM}}$ at Tsirelson 1980 bound $|S| \leq 2\sqrt{2}$; equivalently $S^2 \leq 8$. Experimental verification: Aspect 1982 (Orsay) first loophole-prone test; Hensen 2015 (Delft, Nature) first loophole-free test with NV-center diamonds; ongoing precision tests (Vienna, Caltech, Munich, Hanson). Bell + Tsirelson bound foundational results in quantum foundations. BST T2399 substrate Bell-CHSH: $\text{Tr}(B^2) = 126/16 = 7.875$ EXACT (trace-level integrated-capacity per Calibration #17 Wednesday May 20 refinement: 126/16 is trace-level integrated capacity over 126 active substrate channels, NOT max-eigenvalue framing). Sub-Tsirelson deviation: $\text{Tsirelson}^2 - \text{Tr}(B^2) = 8 - 126/16 = 0.125 = 1/8 = 1/2^N N_c$ — substrate signature at depth $N_c = 3$ below Tsirelson bound. T2469 SCMP framework (Lyra Friday Casey-named #8 STANDING): substrate-mediated observer correlations exhibit 1/8 deviation as Layer 1 operational claim per SCMP bandwidth-bounded observer framework (Vol 5 Ch 10 decoherence + Vol 14 Ch 3 substrate channel capacity). Calibration #17 RESOLVED Thursday via Elie K52a Session 32 rank-1 projector framework $B^2 = (126/16)|\psi_0\rangle\langle\psi_0|$ satisfies BOTH $\text{max}=126/16 + \text{Tr}=126/16$; multi-month question refined to "WHICH substrate-natural $|\psi_0\rangle$ " pending Sessions 30+. SP-30 Substrate Engineering Program Bell experiment design \$300-500K (Vienna/Caltech/Munich/Hanson outreach contacts): high-precision measurement of $|S|^2$ in CHSH inequality at sub-Tsirelson resolution; if measured $|S|^2 < 8$ by exactly 1/8 (0.125) within experimental error $\rightarrow$ BST sub-Tsirelson confirmed; if $|S|^2$ saturates Tsirelson ($|S|^2 \rightarrow 8$ reliably with no 1/8 deviation) $\rightarrow$ BST refuted. Information-theoretic interpretation: Bell-CHSH bounds quantify information-extractable from quantum correlations per Cirelson-Wehner et al. analyses; substrate sub-Tsirelson 1/8 IS substrate-mediation signature distinguishing BST from standard QM channel capacity (Vol 14 Ch 3 substrate-tick capacity bounded by substrate-tick rate). This is BST's sharpest near-term operational falsifier in quantum foundations.

Level 3 (5th-grader accessible): Bell's inequality (1964) shows quantum correlations can be stronger than any classical (local-hidden-variable) theory allows: classical $\text{max } |S| = 2$; QM $\text{max } |S| = 2\sqrt{2} \approx 2.83$ (Tsirelson bound). Experiments since 1982 confirm QM. BST predicts something subtle: quantum correlations should be SLIGHTLY BELOW Tsirelson by exactly $1/8 = 0.125$ — substrate signature at depth $N_c = 3$ (BST integer). High-precision Bell experiment (\$300-500K design at Vienna/Caltech/Munich/Hanson labs) could test this. Information-theoretic interpretation: substrate-mediation IS the 1/8 deviation.

Section 6.1 — Bell 1964 + CHSH 1969 + Tsirelson 1980

Bell inequality: $|S| \leq 2$ for any LHV theory; CHSH form $S = \langle A_1 B_1 \rangle + \langle A_1 B_2 \rangle + \langle A_2 B_1 \rangle - \langle A_2 B_2 \rangle$.

Tsirelson 1980: $|S|_{\text{QM}} \leq 2\sqrt{2}$; $S^2 \leq 8$ (QM bound).

Aspect 1982 + Hensen 2015 + ongoing precision tests.

Section 6.2 — T2399 Substrate Bell-CHSH 126/16 + Cal #17

$\text{Tr}(B^2) = 126/16 = 7.875$ EXACT (trace-level integrated-capacity per Calibration #17 Wednesday refinement).

126/16 = trace-level over 126 active substrate channels.

Section 6.3 — Sub-Tsirelson $1/2^{N_c} = 1/8$ Substrate Signature

$\text{Tsirelson}^2 - \text{Tr}(B^2) = 8 - 126/16 = 0.125 = 1/8 = 1/2^{N_c}$.

Substrate signature at depth $N_c = 3$ below Tsirelson bound.

Section 6.4 — T2469 SCMP Framework Layer 1 (Friday Casey-named #8 STANDING)

Substrate-mediated observer correlations exhibit 1/8 deviation per SCMP bandwidth-bounded observer framework.

Cross-link Vol 5 Ch 10 decoherence + Vol 14 Ch 3 channel capacity.

Section 6.5 — Calibration #17 RESOLVED Thursday

Elie K52a Session 32 rank-1 projector framework: $B^2 = (126/16) |\psi_0\rangle\langle\psi_0|$ satisfies BOTH $\max = 126/16 + \text{Tr} = 126/16$.

Multi-month question refined to “WHICH substrate-natural $|\psi_0\rangle$ ” pending Elie Sessions 30+.

Section 6.6 — SP-30 Bell Experiment \$300-500K Operational Falsifier

SP-30 Substrate Engineering Program Bell experiment design \$300-500K (Vienna/Caltech/Munich/Hanson outreach contacts).

If measured $|S|^2 < 8$ by exactly 1/8 within error $\rightarrow$ BST confirmed. If $|S|^2$ saturates Tsirelson reliably $\rightarrow$ BST refuted.

Section 6.7 — Information-Theoretic Interpretation

Bell-CHSH bounds quantify information-extractable from quantum correlations.

Substrate sub-Tsirelson $1/8$ = substrate-mediation signature distinguishing BST from standard QM channel capacity (Vol 14 Ch 3).

BST’s sharpest near-term operational falsifier in quantum foundations.

Section 6.8 — Honest scope + Connection

- Bell-CHSH + Tsirelson standard [\[7\]](#)
- T2399 + Cal #17 + T2469 substrate framework [\[7\]](#)
- 1/8 sub-Tsirelson BST signature operational falsifier
- Open scope: full operator-level Calibration #17 closure (Elie K52a Sessions 30+ multi-month)

Connection: - Vol 5 Ch 8 Bell-CHSH + Quantum Correlations (pedagogical bridge)
- T2399 + Cal #17 + T2469 SCMP (Friday Casey-named #8) - Vol 14 Ch 3 Shannon channel capacity (substrate-mediation channel) - SP-30 Bell experiment design

— Lyra, Vol 14 Ch 6 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT